

SIVF: Streaming Inverted File Indexing on GPUs

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Abstract

1 Introduction

1.1 Motivation

Real-world vector search applications, such as real-time recommendation and fraud detection, increasingly operate on streaming data where timeliness is critical. These systems require a sliding window model where expired vectors must be evicted as new vectors arrive to maintain bounded memory usage. However, existing GPU-accelerated approximate nearest neighbors (ANN) indices are predominantly optimized for write-once-read-many workloads.

To quantify the gap between insertion and eviction performance, we conducted benchmarks using both the Python (precompiled package through PyPI) and C++ (local compilation from source) interfaces of Faiss [1] on an NVIDIA RTX 6000 GPU with the SIFT1M dataset [2] on the Chameleon testbed [3]. As illustrated in Figure 1, we observe a severe performance asymmetry across both implementations. In the native C++ benchmark, while inserting a batch of 10,000 vectors takes only 28.6 ms due to efficient GPU parallelism, evicting the same number of vectors incurs a latency of 202.2 ms. This constitutes a ~ 7.1 times slowdown, confirming that the bottleneck is fundamental to the index design rather than language binding overhead.

This bottleneck stems from a fundamental architectural limitation in current library designs. Since standard IVF (Inverted File) indices lack support for efficient in-place deletion directly on the GPU, evicting data necessitates a costly CPU-GPU roundtrip, even when using C++. The relevant index structure must be copied back to the host, compacted on the CPU, and re-uploaded to the device. This heavy I/O and CPU burden prevents the system from fully utilizing GPU compute throughput and limits the maximum sustainable ingestion rate.

Acknowledgment

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References

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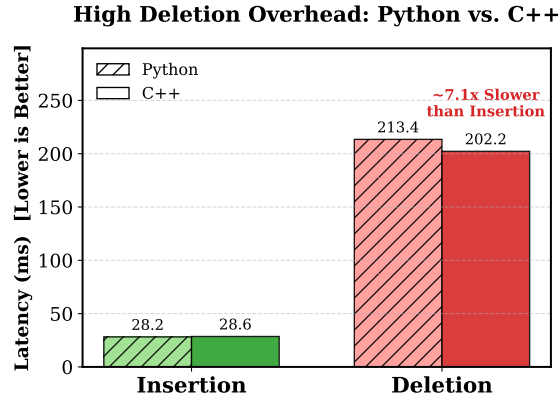


Figure 1: The High Cost of Deletion compared across Python and C++ interfaces. Even using the native C++ implementation, while GPU parallelism accelerates vector insertion (~ 28.6 ms), the lack of in-place GPU eviction support causes deletion latency to surge to ~ 202.2 ms. This represents a ~ 7.1 x slowdown, confirming this asymmetry as the primary bottleneck in streaming scenarios regardless of the language binding.